SELF ASSESSMENT 2 - 4 (Page 70)

1. $P_1 = 2.1 \times 10^3$ kPa $\quad T_1 = 313.15$ K $\quad v_1 = 0.0946$ m^3/kg

But Pv = RT; $\quad R = \dfrac{P_1 v_1}{T_1} = \dfrac{2.1 \times 10^3 \times 0.0946}{313.15} = 0.6344$ kJ/kgK

and $R = \dfrac{\widetilde{R_u}}{\widetilde{M}}$; $\quad$ Hence $\widetilde{M} = \dfrac{\widetilde{R_u}}{R} = \dfrac{8.31447 \text{ kJ/kmolK}}{0.6344 \text{ kJ/kgK}}$

$\widetilde{M} = \mathbf{13.10\ kg/kmol}$

2. Rigid tank with $V = 1.0$ m^3 $\Rightarrow V_1 = V_2 = 1$ m^3

$P_1 = 0.25 \times 10^3$ kPa $\quad V_1 = 1$ m^3 $\quad T_1 = 323.15$ K

$P_2 = ?$ $\quad V_2 = 1$ m^3 $\quad T_2 = 598.15$ K

For $O_2, \widetilde{M_{O_2}} = 32$ kg/kmol $\Rightarrow R_{O_2} = \dfrac{8.31447}{32} = 0.2598$ kJ/kgK

$m = \dfrac{P_1 V_1}{RT_1} = \dfrac{0.25 \times 10^3 \times 1}{0.2598 \times 323.15} = 2.98$ kg

$P_2 = \dfrac{mRT_1}{V_1} = \dfrac{2.98 \text{ kg} \times 0.2598 \text{ kJ/kgK} \times 598.15\text{K}}{1 \text{ m}^3}$

$P_2 = 463.09 \dfrac{\text{kJ}}{\text{m}^3} \left(\dfrac{\text{kNm}}{\text{m}^3} = \dfrac{\text{kN}}{\text{m}^2} \right) = 463.09$ kPa $= \mathbf{0.463\ MPa}$

3. Air in a rigid container

$R_{air} = 0.287 \dfrac{\text{kJ}}{\text{kgK}}$; $P_1 = 0.1 \times 10^3$ kPa; $\quad \rho = 1.0 \dfrac{\text{kg}}{\text{m}^3}$; $m = 453.6$ kg

$P_1 = \rho R T_1$; $\quad T_1 = \dfrac{P_1}{\rho R} = \dfrac{0.1 \times 10^3 \text{ kN/m}^3}{1.0 \dfrac{\text{kg}}{\text{m}^3} \times 0.287 \dfrac{\text{kJ}}{\text{kgK}}} = 348.43$ K

$\rho = \dfrac{m}{V}$; $\quad V = \dfrac{m}{\rho} = \dfrac{453.6 \text{ kg}}{1.0 \dfrac{\text{kg}}{\text{m}^3}} = \mathbf{453.6\ m^3}$

4. Sealed volume of 2 cm^3 $= 2 \times 10^{-6}$ m^3 contains saturated water

$P_1 = 0.10 \times 10^3$ kPa $= 100$ kPa

(a) mass $= 0.10$ g

At final state $v = \dfrac{V}{m} = \dfrac{2 \times 10^{-6} \text{ m}^3}{0.1 \times 10^{-3} \text{ kg}} = 2 \times 10^{-2} \text{m}^3/\text{kg}$

From table at P = 100 kPa, $\quad v_g = 1.6941$ m^3/kg

(b) mass $= 1\text{g} = 1 \times 10^{-3}$ kg $\quad v_2 = \dfrac{2 \times 10^{-6} \text{m}^3}{10^{-3}} = 2 \times 10^{-3} = 0.002 \dfrac{\text{m}^3}{\text{kg}}$

(a) $0.02\,\frac{m^3}{kg}$ as single phase water is between

v	T_{sat}
0.020489	303.35
0.02	T?
0.018	311.00

$$\frac{T_{sat} - 303.35}{311 - 303.35} = \frac{0.02 - 0.020489}{0.018 - 0.020489}$$

$T_{sat} = \mathbf{304.85}$ ℃

(b) $v_2 = 0.002\ m^3/kg$ as angle phase water

v	T_{sat}
0.001926	361.47
0.02	T?
0.002038	365.75

$$T = 341.47\ ℃ + \frac{0.002 - 0.001926}{0.002038 - 0.001926} \times (365.75 - 361.47)$$

$T = \mathbf{364.30}$ ℃

5. Initial state 1 kg of H_2O at 200℃, $P = 800$ kPa

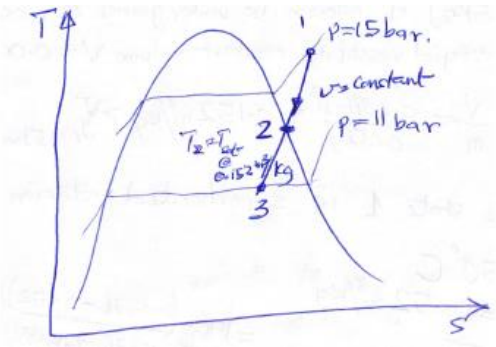
At $P_1 = 800$ kPa and $T_1 = 200$℃, $v_1 = 0.26088\ m^3/kg$

State 2 $P_2 = 800$ kPa at $T_2 = 170.41$ ℃, $v_2 = v_{f_{@800kPa}} = 0.001115\,\frac{m^3}{kg}$

$$W_{12} = -P(v_2 - v_1) = -800(0.001115 - 0.26088) = \mathbf{207.8\ kJ/kg}$$

6. **Answer:** The weight of the pan and its contents is much less than the reaction force at the pan-lid interface. Therefore, the pan will move up with the lid when the person attempts to open by lifting the lid up.

7. Fixed mass (0.05 kg) of stem is undergoing series of processes in a rigid vessel of constant volume V = 0.0076 m^3.



State 1 $v_1 = \frac{V}{m} = \frac{0.0076\ m^3}{0.05\ kg} = 0.152\ m^3/kg > v_{g_{@15\ bar}}$

Hence state 1 is superheated steam and $T_1 = 250\ ^\circ C$

$v_1 = v_2 = v_3 = 0.152\ m^3/kg$

At state 2, $T_2 = T_{sat@v_2=0.152\ m^3/kg} = 190 + \dfrac{0.15636 - 0.152}{0.15636 - 0.14089}(195 - 190)$

$T_2 = 191.4\ ^\circ C$

At state 3, $P_3 = 11$ bar and $v_3 = 0.152\ m^3/kg$

but $v_{f_{@11\ bar}} < v_3 < v_{g_{@\ 11\ bar}}$

$$X_3 = \frac{v_3 - v_{f_{@11\ bar}}}{v_{g_{@\ 11\ bar}}} = \frac{(0.152 - 0.001133)}{(0.177 - 0.001133)} = 0.857$$

$u_1 = 2697\ kJ/kg$

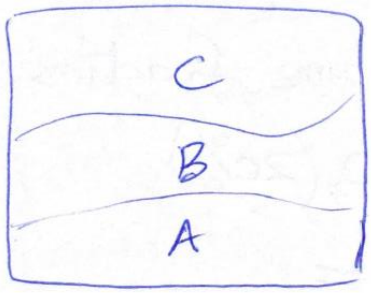
$u_3 = u_f + x_3 u_{fg} = 780 + 0.857(2586 - 780) = 2327.7\ kJ/kg$

$Q_{13} = m(u_3 - u_1)$ since $W_{13} = 0$

$Q_{13} = 0.05(2327.7 - 2697) = -18.5\ kJ$

Hence, the heat rejected is 18.5 kJ

8. Three gases in a closed vessel with $P_{Total} = 300\ kN/m^2$



Generally, for a gas $P_{mix}V_{mix} = \tilde{n}_{mix}\widetilde{R_u}T_{mix}$

For constituent A, $P_A V_{mix} = \tilde{n}_A \widetilde{R_u} T_{mix}$

For constituent B, $P_B V_{mix} = \tilde{n}_B \widetilde{R_u} T_{mix}$

For constituent C, $P_C V_{mix} = \tilde{n}_C \widetilde{R_u} T_{mix}$

$$\frac{P_A}{P_{mix}} = \frac{n_A}{n_B};\ \frac{P_B}{P_A} = \frac{n_B}{n_A} \Rightarrow P_B = \frac{n_B}{n_A}P_A \text{ and } P_C = \frac{n_C}{n_A}P_A$$

$$P_{mix} = P_A\left[1 + \frac{n_B}{n_A} + \frac{n_C}{n_A}\right]$$

But $\dfrac{V_A}{V_{mix}} : \dfrac{V_B}{V_{mix}} : \dfrac{V_B}{V_{mix}} = \dfrac{n_A}{n_{mix}} : \dfrac{n_B}{n_{mix}} : \dfrac{n_C}{n_{mix}} = \dfrac{5}{20} : \dfrac{8}{20} : \dfrac{7}{20}$

$\therefore\ P_A = \dfrac{5}{20} \times 300\ kN/m^2 = 75\ kN/m^2$

$P_B = \dfrac{8}{20} \times 300\ kN/m^2 = 120\ kN/m^2$

$P_C = \dfrac{7}{20} \times 300\ kN/m^2 = 105\ kN/m^2$

9. Gas mixture with given volume fractions

O_2 (10%) ; N_2 (70%) ; CO_2 (20%)

Total pressure = 1 bar

$$P_{O_2} = \frac{10}{100} \times 1 \text{ bar} = 0.1\text{bar} = 100 \text{ mm bar}$$

$$P_{N_2} = \frac{70}{100} \times 1 \text{ bar} = 0.7\text{bar} = 700 \text{ mm bar}$$

$$P_{CO_2} = \frac{20}{100} \times 1 \text{ bar} = 0.2\text{bar} = 200 \text{ mm bar}$$

10. Volume $_{\text{mixture}} = 320 \text{ dm}^3$

Constituents CO_2 (0.4 kg); CO (0.04 kg); O_2 (0.12 kg)

$T_{mix} = 18\ ℃ = 291 \text{ K}$

$\breve{M}_{CO_2} = 44 \text{ kg/kmol}$ $\breve{M}_{CO} = 28 \text{ kg/kmol}$ $\breve{M}_{O_2} = 32 \text{ kg/kmol}$

$\breve{R}_u = 8.31447 \text{ kJ/kmolK}$

$$R_{CO_2} = \frac{8.31447}{44} = 0.189 \text{ kJ/kgK}$$

$$R_{CO} = \frac{8.31447}{28} = 0.297 \text{ kJ/kgK}$$

$$R_{O_2} = \frac{8.31447}{32} = 0.260 \text{ kJ/kgK}$$

$$n_{CO_2} = \frac{0.4}{44} = 0.00909$$

$$n_{CO} = \frac{0.04}{28} = 0.00143$$

$$n_{O_2} = \frac{0.12}{32} = 0.00375$$

$$n_{mix} = 0.01427$$

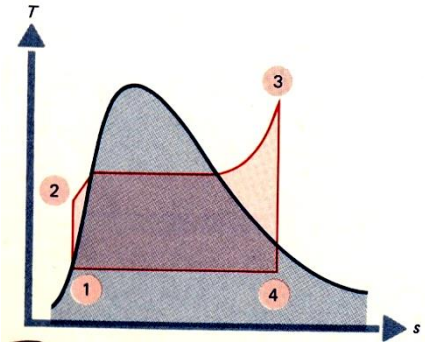
$$P_{CO_2} = \frac{0.00909 \times 8.31447 \times 291}{320 \times 10^{-3} \text{ m}^3} = 68.7 \text{ kN/m}^2$$

$$P_{CO} = \frac{0.00143 \times 8.31447 \times 291}{320 \times 10^{-3} \text{ m}^3} = 10.8 \text{ kN/m}^2$$

$$P_{O_2} = \frac{0.00375 \times 8.31447 \times 291}{320 \times 10^{-3} \text{ m}^3} = 28.4 \text{ kN/m}^2$$

$$P_{Total} = 68.7 + 10.8 + 28.4 = 107.9 = 108 \text{ kN/m}^2$$

SELF ASSESSMENT 4 - 4 (Page 79)



1. $|\dot{W}_t| = 1500\ \text{kW} \qquad \dot{m}_s = 1.5\ \text{kg/s}$

$|\dot{W}_t| = \dot{m}_s(h_3 - h_4) = 1500\ \text{kW}$

But $h_4 = h_{f@\,200\,\text{kPa}} + x_4 h_{fg@\,200\,\text{kPa}} = 504.71 + 0.98(2201.6)$

$h_4 = 2661.84\ \text{kJ/kg}$

$$h_3 - h_4 = \frac{1500\ \text{kJ/kg}}{1.5\ \text{kg/s}} = 1000\ \text{kJ/kg}$$

$\therefore h_3 = 1000 + 2661.84 = 3661.84\ \text{kJ/kg}$

$$T_3 = 500 + \left(\frac{3661.84 - 344.6}{3674 - 344.6}\right)(600 - 500) = 594.6\ ℃$$

State 3 is superheated since $h_3 > h_{g\,@\,4\text{MPa}}$ and $T_3 = 594.6\ ℃$

$|w_p| = h_2 - h_1 = v_{f1}(P_2 - P_1) = 0.001061(4000 - 200) = 4.03\ \text{kJ/kg}$

$|\dot{W}_p| = \dot{m}_s|w_p| = 1.5\ \text{kg/s} \times 4.03\ \text{kJ/kg} = 6.045\ \text{kW} \cong 6\ \text{kW}$

Hence, $|\dot{W}_{net}| = |\dot{W}_t| - |\dot{W}_p| = 1500\ \text{kW} - 6\ \text{kW} = 1494\ \text{kW}$

$h_2 = 4.03 + h_1 = 4.03 = 504.71 = 508.74\ \text{kJ/kg}$

$q_{in} = h_3 - h_2 = 3661.84 - 508.74 = 3153.1\ \text{kJ/kg}$

$\dot{Q}_{in} = \dot{m}_s q_{in} = 1.5\ \text{kg/s} \times 3153.1\ \text{kJ/kg} = 4729.65\ \text{kW}$

$$\eta_{th} = \frac{|\dot{W}_{net}|}{\dot{Q}_{in}} = \frac{1494}{4729.65} \times 100\ \% = \mathbf{31.6}\ \%$$

2. $T_{cooling} = 20\ ℃; \qquad T_{condenser} \geq 20 + 15 = 35\ ℃$

$P_{sat@\,35\,℃} = 5.63\ \text{kPa} < P_{condenser} = 10\ \text{kPa}$

Hence, it is possible to maintain a pressure of 10 kPa in a condenser that is being cooled by river water entering at 20 ℃. $T_{condenser}$ is 45.81 ℃ and that's enough temperature difference for heat transfer.

TYPICAL UNIT 4 CALCULATION MCQ'S IN ME 265 - DETAILED SOLUTIONS

1. $T = T_{sat@350\,\text{kPa}} = 138.86\ ℃ \qquad v = v_{g@350\,\text{kPa}} = 0.52422\ \text{m}^3/\text{kg}$

$$m = \frac{V}{v} = \frac{0.6\ \text{m}^3}{0.52422\ \text{m}^3/\text{kg}} = \mathbf{0.114\ kg}$$

2. $P = P_{sat@90\,℃} = 70.183\ \text{kPa}$

But at 90 ℃, $v_f = 0.001036\ \text{m}^3/\text{kg}$ and $v_g = 2.3593\ \text{m}^3/\text{kg}$

Also, $V = V_f + V_g = m_f v_f + m_g v_g = 8 \times 0.001036 + 2 \times 2.3593 = \mathbf{4.73\ m^3}$

3. $v = \frac{V}{m} = \frac{0.08}{4} = 0.02\ m^3/kg$

But at 160 kPa, $v_f = 0.0007437\ m^3/kg$ and $v_g = 0.12348\ m^3/kg$

But, $v_f < v < v_g$ hence, $x = \frac{v - v_f}{v_{fg}} = \frac{0.02 - 0.0007437}{0.12348 - 0.0007437} = \mathbf{0.157}$

4. $v = {RT}/{P} = \frac{0.0815\frac{kJ}{kgK} \times 323.15}{1000\ kPa} = \mathbf{0.026369}$

5. $P = P_{sat_{@90\,℃}} = 70.183\ kPa$ $v = v_{f_{@90℃}} = 0.0010369\ m^3/kg$

Total volume of tank $= mv_f = 50\ kg \times 0.001036\ m^3/kg = \mathbf{0.0518\ m^3}$

6. $P = P_{sat_{@90\,℃}} = 70.183\ kPa$

7. $v = \frac{V}{m} = \frac{0.08}{4} = 0.02\ m^3/kg$

but at 160 kPa, $v_f = 0.0007437\ m^3/kg$ and $v_g = 0.12348\ m^3/kg$

since $v_f < v < v_g$, $T = T_{sat_{@160\ kPa}} = \mathbf{-15.60}\ ℃$

8. $P_3 = 3\ MPa$, $T_3 = 600\ ℃$, $h_3 = 3682.8\frac{kJ}{kg}$, $s_3 = s_4 = 7.5103$

$P_4 = 10\ kPa$, $s_g = 8.1488 < s_4$ (saturated liquid vapour mixture)

$s_f = 0.6492\ kJ/kgK$ $s_{fg} = 7.4996\ kJ/kgK$

$\therefore x_4 = \frac{7.5103 - 0.6494}{7.4996} = 0.915$; $h_4 = 191.81 + 0.915(2392.1) = 2380.6\ kJ/kg\ K$

$q_{in} = h_3 - h_1 = 3491\ kJ/kgK$ $q_{out} = h_4 - h_1 = 2380.6 - 191.81 = 2188.8\ kJ/kgK$

$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{2188.8}{3491} = 0.3730 = \mathbf{37.3}\ \%$

9. $P_3 = 5\ MPa, 600\ ℃$ $s_3 = 7.2605\ kJ/kgK$ $x_4 = \frac{7.2605 - 0.6492}{7.4996} = 0.882$

$\therefore 1 - x_4 = 1 - 0.882 = 0.118 = 11.8\ \% \cong \mathbf{12}\ \%$

10. $h_2 = h_1 = h_{f_{@10\ kPa}} = 191.81\ kJ/kgK$

$h_3 = h_{@_{5MPa,600℃}} = 3666.9\ kJ/kgK$

$s_3 = s_{@_{5MPa,600℃}} = 7.2605\ kJ/kgK$

$x_4 = \frac{7.2605 - 0.6492}{7.4996} = 0.882$

$h_4 = 191.81 + 0.882(2392.1) = 2301.8\ kJ/kg\ K$

$\dot{m} = \frac{300\ kg/s}{(3666.9 - 191.81)} = 0.0863\ kg/s$

$|\dot{W}| = \dot{m}(h_3 - h_4) = 0.0863(3666.9 - 2301.8) \cong \mathbf{118\ kW}$